

The Hong Kong Polytechnic University
Department of Computing

COMP5926 Metaverse Project I
Proposal

Digital Twin for Elderly Home Service

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1. Abstract

Population ageing and growing number of elderly aging in place have increased the risk of “delayed discovery”. If a fall, sudden illness, or other incident occurs and goes unnoticed, the elderly may die and lie undiscovered for a long period. Existing in-home monitoring solutions often rely solely on physiological signals collected by wearables, which may be insufficient to distinguish incidents. Camera monitoring may not be acceptable due to privacy concerns and can raise ethic and legal issues.

This proposal presents a privacy-friendly Digital Twin for elderly home service, integrates wearable sensing with non-visual ambient sensing and high-precision indoor localization. The system constructs in individualized digital twin of daily life patterns, visualizing real-time status. Dashboards and activity heatmap are used for visualizing the collected data. Machine learning models extracts features to established a personalized baseline, then be applied in deviation detection. Alarm mechanism is designed to confirm their status and escalates notification to family members or medical services. In addition, a voice/text AI assistant provide check-ins, non-diagnostic advice can meet daily needs of the elderly.

Expected outcomes include a prototype system with digital twin for real-time event visualization and personalized abnormal behavior detection with alert. This project aims to implement and evaluate camera-free, context-aware in-home monitoring and contribute a deployable framework for safer aging in place

Keywords: Digital Twin, Elderly Home Service, Anomaly Detection

2. Introduction

Population aging is accelerating in the worldwide. As younger family members often work or study away from home, many elderly live alone or only with a spouse. Aging in place offers autonomy and reduces the pressure on institutional care compared to reside in elder care facilities, but the risk in the event of acute illness or functional decline may increases. When incidents like falls, loss of consciousness, or exacerbation of chronic disease occur, it will greatly affect their mobility and safety. Without notifying other members, it will put their life in danger, and lead to a tragedy, the elderly found with no vital signs. Significantly, such incidents not occur in public areas but in private, confined spaces like bedrooms or bathrooms, where they are unlikely to get noticed.

Recent news reports about the deaths of Academy Award-winning actor Gene Hackman and his wife highlights a critical problem that even for affluent families, when elderly live in private and confined area with limited communication with outside, their condition may difficult to monitor, leading to the risk of being undiscovered for hours or longer during life-threatening events. Such cases alert us that the key to guarantee the safety of elderly who aging in place, is not solely monitoring their vital signs, but in identifying abnormal behaviors early and aiding them when a critical condition occurs.

Current health monitoring technologies are highly developed, physical signals such as heart rate, pulse, oxygen saturation, blood pressure or sleep condition, can be monitored via wearable or other sensors. However, only relying on physiological signals may insufficient in indicating elderly person's status. For example, they may still have pulse while siting on the sofa for over 4 hours without other movements, lying in bed far beyond their usual routine, or even fall in the bathroom and lose their mobility. These kinds of crisis cannot be determined by single parameter. Therefore, integrating in-home positional and mobility context with physiological signals is necessary for reliable detection of emergencies.

Meanwhile, in-home monitor also leads to the privacy concern. While camera can capture clear images, provide rich information that can reveal elder person's status, but it is often unacceptable in private spaces and raises ethic and legal issues. Thus, how to locate the elderly continuously and monitor their current movements by using privacy-friendly sensors deploy in their house is the key problem that need to resolve.

Base on this, this proposal suggests a solution that integrates Digital Twin technology into elderly home services, combines with AI technique for analysis and response. Wearable device such as smart watch to collect elderly's physiological signals. Privacy-friendly ambient sensors which can detect pressure and position, can be installed in furniture to provide non-visual evidence of presence, posture, and movement. The Digital Twin system constructs a personalized life

pattern of the elderly, consistently records and visualizes their activities, and alerts their families when abnormal behavior is detected or emergencies occur. The AI agent enables voice-based interaction for check-ins and provides non-diagnostic health information, thereby reduces operation difficulty for the elderly.

3. Objectives

This proposal aims to develop a privacy-preserving, in-home health elderly monitoring system that integrates wearable sensing for continuous physical signal detection, non-visual ambient sensing for movement tracking, and a digital twin representation of life patterns. The system combines IoT and AI technologies to detect emergencies and provide assistance for elderly in-home health monitoring. The objectives of this proposal are as follows:

Physiological Signal Detection and Logging: Collect elderly user's physiological signals via wearable devices, provide visualization for observation and maintain secure records for medical services.

Digital Twin Construction of Life Patterns: Construct an individualized Digital Twin that models elderly daily routines. Use multiple privacy-friendly ambient sensors to detect an elderly individual's presence and movements, provide visualization such as heatmap for their daily activities. Utilizing individual's routine for abnormal behaviors detection algorithms.

Abnormal Behavior Detection and Alert: Based on the elderly individual's routine, developed and evaluate algorithms that can detects deviations. Upon detecting abnormal behavior, the system first notifies the individual. They can cancel this safeguard notification if it is a false alarm. If the user does not respond, the alert is then escalated to their family or medical services.

AI Assistant for Check-In and Medical Support: Implement a voice/text AI assistant that supports check-ins and provides non-diagnostic health information and incident guidance, and can initiate alerts for aid upon individual request.

4. Methodology

4.1 Data Collection and Analysis

To detect abnormal behaviors, the entire procedure involves collecting and pre-processing raw data, extracting features for machine learning models, developing baselines for machine learning, detecting anomalies, and iterating models over the complete lifecycle of a digital twin representing the elderly's physiological signals. The full process can be divided into three primary sub-processes:



Fig 1. Elderly with smart wearable device

4.1.1 Raw Data Preprocessing

To collect the physiological (heart rate, blood pressure) and positioning data of the elderly user, the system collects signals from a smart bracelet or other wearable devices and visualizes this data so it can be viewed in real-time. The data collected is deidentified before being saved securely for the use of medical professionals. AI will clean up the collected data (remove input errors from devices or remove invalid entries that are influenced by environmental interference of devices and do not represent the elderly user) and standardize the data (convert blood pressure to the same unit of measure and format the location of the signal to match the location on the map of the house (i.e., converting the location of the bathroom to the electronic map of the house)) to protect the privacy of the elderly user while providing consistency across recorded data (Jeong et al. , 2017).

4.1.2 Feature Extraction and Developing Baseline Digital Twin Models

From the data collected from both the wearables and the ambient sensors (detecting presence, posture, and movement) of the elderly, algorithm will extract two types of features that will be used to identify the normal baseline of each elderly user: physiological indicators of the user (e.g., changes in heart rate or blood pressure over the last 24 hours and circadian rhythms) and behavioral indicators of the user, including time spent in a specific area of the house, frequency of activity

performed, and time-of-day spent on activities. Using these four identifiers, algorithm will create a baseline for the elderly user using not less than seven days and no more than 14 days of continuous data collected over the previous seven to 14 days.

4.1.3 Detecting Abnormal Behavior and Iterating AI Models

Detecting algorithm will analyze the daily baseline data collected on the elderly user to develop and evaluate the existence of anomalies (deviations from the normal baseline). It will perform multi-dimensional fusion of the physiological and positional data to detect the abnormal behavior (e.g., spending more than 30 minutes in the bathroom and having a heart rate greater than 100 beats per minute is classified as a high-risk abnormal behavior and sitting on the couch for more than four hours and having blood pressure with little or no deviation from the normal baseline is classified as a low-risk abnormal behavior). When algorithm detects an abnormal behavior, the person being monitored will be notified that there was an abnormal behavior detected. The person being monitored can cancel the alarm notification if the notification was triggered by a false alarm. If the alarm is not canceled or is not acknowledged within a specified time period, the system will escalate the notification to the elderly's family or the medical professionals.

As part of the normal operation of the system, the system will maintain a record of the match of the how the abnormal behavior occurred and the actual behavior that the elderly user exhibited at that particular point in time, thus allowing the AI models to continuously refine the baseline thresholds and reduce the false alarm rates.

4.2 Digital Twin Construction of Life Patterns

The Digital Twin component serves as the central interface for monitoring, bridging the physical home environment with a virtual representation. This system is built using NVIDIA Isaac Sim. Unlike traditional monitoring systems which provide intrusive video streams, our solution reconstructs 3D virtual avatars of elderly users. This approach uses general 3D avatar visualization to display their real-time location, approximate posture, and health status, thereby protecting users' dignity and privacy.

4.2.1. Virtual Environment Modeling

The Digital Twin will only be a true representation of the actual testbed if it has an accurate one-to-one correspondence with the physical environment. To achieve this, we will use a hybrid modeling approach in conjunction with manual calibrations based on images taken from LiDAR scans to build a three-dimensional high-fidelity model of the home of the senior subject. Items in the home will be documented for layout and representation within Isaac Sim using USD.

And some key furniture assets such as beds and sofas. These will have physical colliders and sensor areas marked in the virtual world, also correspond to the IDs of physical pressure sensors

installed in the real world. This ensures a logical mapping between physical sensor activation and virtual environmental context.

4.2.2. Avatar Kinematics and Posture Reconstruction

Because our system does not utilize cameras to track bone connections, we cannot use conventional motion capture methods. As an alternative, we suggest using a state machine-based Sensor Fusion (SM-SF) to deduce and display the user's actions. The 3D avatar will be generated by fusing measurements from the spatial sensors, pressure pads, and the Apple Watch.

Real-Time Localization: The first area of focus is to synchronize the coordinates of the user. The user's (x,y) coordinates are reported by Ultra-Wideband (UWB) or millimeter Wave Radar sensors. The coordinates will then be transformed using a transformation matrix to map the coordinates into Isaac Sim's world coordinate system. This way, we would create a coordinate system within Isaac Sim.

The second area of focus is to smooth out the avatar's movement. The avatar may appear to be "jittering" due to sensor noise. A Kalman Filter will be used to predict the user's trajectory and to smooth the avatar movement so the visualization appears seamless and natural when the user moves from one room to another.

Inference-Based Posture Visualization: In order to successfully show the activity of an elderly user (i.e., Standing, Sitting or Lying Down) with no visual privacy infringement by the use of cameras using a logical-based animation controller/scheme, the animation controller operates in accordance with the logic of a state machine, transitioning the 3D virtual human from one predefined state to another based on predefined sensor fusion thresholds (Debard et al., 2015).

The first operation of the system is to distinguish between an Active and Inactive state. The Standing and Walking states will be indicated when there is a height detected from the mmWave radar exceeding 1.4m, combined with a horizontal velocity change in x,y coordinate system exceeding 0.1m/s. If both of these conditions are met, the virtual avatar will perform either a "Walking" or "Standing Indefinitely" animation from its current position. The Sitting State will be indicated when the mmWave radar detects a change in height to a range of 1.0m to 1.2m. This indication needs to be provided together with the same timestamp from the sofa or chair pressure sensor to confirm Environmental Context. Once the Environmental Context is confirmed the virtual avatar will immediately assume a seated position on the Virtual Furniture.

In the scenarios of Rest or Emergency, the logic also makes use of wearable telemetry data. The user will be considered to have transitioned to the Lying Down state when there is a height below 0.8m detected by the bed pressure sensor. In addition to the height data, the system will use the gyroscopic data from the Apple Watch to confirm that the user's body position is horizontally oriented prior to triggering the virtual avatar's 'Sleeping' animation.

Most importantly, a Fall is identified through a specific combination of negative and positive signals: if the radar detects a rapid height drop to near floor level which shown in coordinates approximately 0 meters without any activation from furniture pressure sensors, and this coincides with the Apple Watch's high-G impact signal. The system will recognize a fall (Casilari & Oviedo-Jiménez, 2015). The avatar will immediately display a collapsed pose to visually alert caregivers of the emergency.

4.2.3. Health Status and Historical Visualization

Beyond physical actions, digital twins will visualize invisible physiological data, providing a more comprehensive view of users' overall health status.

Vital Sign Dashboard: A "billboard," or digital interface associated with the 3D digital model, will appear on either side of your avatar. It will include all of the real-time telemetry data streams as provided by the Apple Watch (e.g., heart rate in beats per minute, and blood oxygen saturation level). The system uses a low-latency data pipeline to map these real-time data streams into the virtual space. Thus, as a user moves throughout that virtual space, real-time updates of vital signs and the user's location and position in the virtual space will allow the caregiver to monitor the vital signs of their patient in a more dynamic, intuitive, and contextually aware manner than traditional numeric measurement methods of vital signs monitoring.

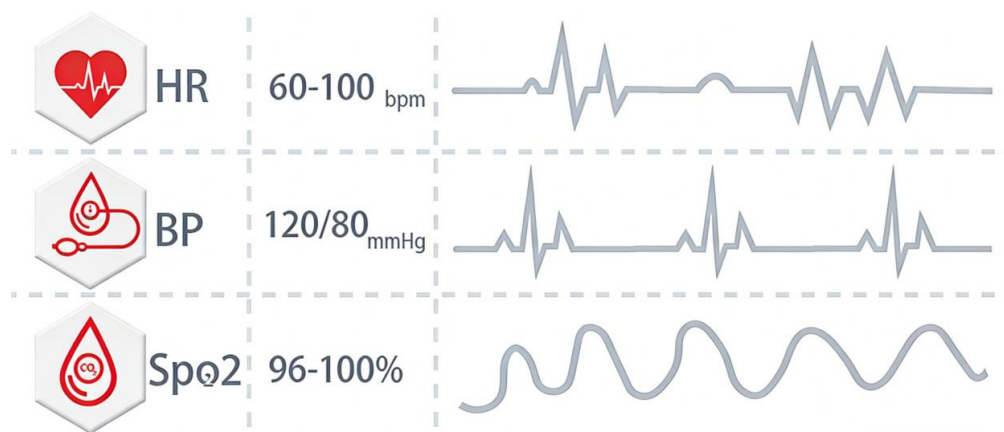


Fig 2. Example of Vital Sign Dashboard of Normal Human

Activity Heatmaps: To conduct an analysis of long-term trends with this system, it collects location-based data that accumulates over time and displays heat maps of this data on top of virtual building floor plans. These heat maps will be used to depict habitual behavioral patterns of elderly individuals, based on where they spend extended amounts of time, and provide caregivers with insight into a possible reduction in mobility or increased risk for depression in elderly individuals who are spending excessive amounts of time in sedentary environments. The data collected from the heatmap can be further analyzed by performing time-based stratification of the data, creating

trend charts to help caregivers or healthcare teams develop an understanding of gradual shifts in behavioral patterns rather than relying on isolated observations of behavioral changes.

4.3 High-Precision Indoor Localization System

Using video cameras for tracking systems has provided helpful options for developing monitoring systems that allow seniors to be cared for while retaining their dignity. However, we prefer to obtain information about a person's location through the use of physical means and not video cameras. Many types of indoor positioning systems rely on signal strength based on RSSI (Received Signal Strength Indicator) including Wi-Fi and Bluetooth, but these systems provide too imprecise data for our needs, since a range of only 0.5 meters could indicate whether or not a senior person is safe (lying on the bed) or incapacitated on the floor next to the bed. Therefore, we have selected to implement an ultra-wideband (UWB) system, which uses the fact that usable signal pulses can yield measurements with a precision of one to two centimeters and arrive at the measuring device(s) within a time delay measured in nanoseconds.

4.3.1 UWB Ranging Principles and Coordinate Calculation

The installation consists of fixed anchor points (i.e., base stations mounted on ceilings or walls) and mobile tags worn by the individual. The operation is as follows: the mobile tag sends a signal pulse, the fixed anchor points record the time they receive the signal from the mobile tag. By crunching the speed of light against the signal's Time-of-Flight, the system figures out the exact distance between the user and each Anchor. (Alarifi et al., 2016)

To convert the aforementioned calculated distances of the mobile tag to fixed anchor point coordinates to the two dimensional map of the coordinates of the area that we are working in, we will use a technique known as trilateration. To determine the location of the mobile tag in real time, we will use at least three fixed anchors in proximity to the area that we are tracking and run a least squares regression on both the different fixed anchor locations and the distance to calculate the mobile tag's location in the instant of calculation.

4.3.2 Hardware Deployment Strategy

The greatest challenge with indoor positioning systems is not with the sensors themselves but with the location of the sensors that are to be installed. Typically, the installation of sensors is a guess as to where to install the sensors. After installing, the person who installed the sensors discovers that the signal path has been obstructed by furniture or some other obstruction and cannot reliably register a signal between the mobile tag to all of the fixed anchor points resulting in a large error due to non-direct line modes. To improve the process as described above, we will recreate a Digital Twin of the home of the monitored senior to allow for replication of their house in a virtual 3D environment. A Slime Mold Algorithm (SMA) will be conducted on the digital twin of the

monitored senior's home, simulating 1000's of potential configurations for the location of the base stations. Once the configurations have simulated, evaluations will be conducted to determine the optimum configurations for the base stations with respect to signal coverage and total potential error incurred prior to drilling any holes in the real walls of the home. (Lou et al., 2022)

4.3.3 Digital Twin-Based Error Mitigation

Even with optimal placement of the anchors, physical reality is fraught with interference. For example, refrigerators/cabinets obstructing signals will result in the mobile tag's signal being reported by the system as being further away than the actual distance exists between the mobile tag and the anchors because of the obstruction (i.e., NON-Line-of-Sight Errors). Asynchronous with the allowance of static error, we will employ AI and train it to produce a new distance measurement allowing us the capability to more accurately determine the true location of the mobile tag to the highest resolution.

4.4 AI Voice Assistant

An intelligent voice/text AI assistant plays a key role as an auxiliary component of the system. In addition to basic check-in functions and non-diagnostic health information and incident support, the assistant can also trigger alerts for assistance at the request of the elderly. To meet the daily needs of the elderly and improve their system usability, the AI voice assistant has three important functional components described below (Pradhan et al., 2020).

4.4.1 Voice Operation Guide for Mobile Phones

This feature gives elderly users a step-by-step voice guide for how to operate their mobile phones through high-frequency usage, such as making phone calls, sending WeChat messages and checking the weather, both of which are frequently used by the elderly. To accommodate the elderly's cognitive abilities and prevent information overload, once the elderly have completed each operational step, the AI will provide a voice confirmation that they have and can move on to the next step once they understand the previous one.

4.4.2 Time-of-day Needs Check-In and Physical Condition Assessment

The AI voice assistant will call the user, via voice three times a day (9:00 a.m., 2:00 p.m., and 7:00 p.m.), to check on the user's physical condition (e.g., "Did you experience any dizziness today, chest sensation, etc.?") as well as to check on his/her daily needs (e.g., "Do you need help contacting your family or getting an item repaired?") (Wang et al., 2024). The system will automatically record the user's responses, categorize/ organize his/her needs, and synchronize the data to the family's terminal they can monitor their family members' activities in real time.

4.4.3 Daily & Weekly Health Reports

Each day at 9:00 p.m., the AI will create a daily health report from the historical data and activity procedure data it collected for the previous day. Weekly, the AI will generate a summary report from the daily health reports to that point. Reports include core physiological measurement trends; daily activity totals; unusual occurrences; and recommendations for improving the individual's health. Reports may be generated in two forms; as a written document in detail that can be sent to the family terminal and/or verbally via a voice broadcast of the semi-daily health report using simplified and easy to understand vocabulary and language to provide the most important health summary.

5. Conclusion

In-home elderly service faces a “delayed discovery” risk: incidents such as falls, sudden illness, or loss of mobility occur in private spaces and may remain unnoticed for long periods. Current monitoring solutions put heavy weight on physiological signals, yet these signals alone are insufficient for reliable detection, and camera monitoring may raise ethic and legal issues.

This proposal suggests a privacy-friendly Digital Twin solution for in-home elderly service. It combines wearable physiological sensing with non-visual ambient sensing to utilize both data for modelling an elderly individual’s daily routine. By applying machine learning algorithms to detect abnormal behaviors, the system can alert family or medical services at an early stage to save their life. In addition, an AI agent with voice/text interaction can be integrated to lower the barrier for elderly to seek assistance, and non-diagnostic medical advice to help conditions from deteriorating.

The expected outcomes include a functional prototype, a Digital Twin representation for elderly real time status and life pattern visualization, dashboard for physiological data, abnormal behaviors detection algorithms and alarm system for emergency. This project seeks to contribute a practical, privacy friendly approach for safer aging in place, and may extend to more scenario for elderly service.

6. Project Development Schedule

The development procedure is divided into two phases, the Proof of Concept (POC) version and the Minimum Viable Product (MVP) version, Further details are as follow:

POC Phase: (12 Jan – 20 Mar)

Week 1- Week 4: Proposal and literature review

Week 5 -Week 10: implementation

Wearable data collection and visualization: implement physiological signal acquisition (e.g., heart rate, blood pressure) and real-time dashboard visualization.

Single-room localization: implement zone-level localization using one or more ambient sensors.

2D Digital Twin visualization: build a 2D floorplan-based Digital Twin to display real-time location and event logs.

Abnormal behavior detection (v1): implement initial detection for a small set of deviations such as immobility, and connect it to a basic alert workflow.

MVP Phase: (20 Mar – 10 Jul)

Improved abnormal behavior detection (v2): Refine baseline modeling and multimodal fusion (physiology signals and location context). Expand the deviation set (more abnormal patterns).

Multi-room coarse localization: Extend localization coverage from a single room to multiple rooms using additional ambient sensors.

3D Digital Twin visualization: Upgrade from 2D to 3D visualization to provide clearer activity descriptions, provide animation.

AI assistant: implement a voice/text assistant to provide non-diagnostic guidance.

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